

MODUL VI

Kolaborasi dan Implementasi Git

Tujuan

- ◆ Mahasiswa paham membuat dan mengelola Issue di GitHub.
- ◆ Mahasiswa mengaplikasikan feature-branch workflow: membuat branch dari Issue, commit terstruktur, push, open PR.
- ◆ Mahasiswa dapat membuat PR menggunakan template dan melakukan peer code review.
- ◆ Mahasiswa dapat meng-merge PR setelah approve dan menutup Issue.
- ◆ Mahasiswa menambah fitur filter items by category (kasus sederhana) dan mendokumentasikannya.

Materi

1. Branching Strategy
 - 1.1. Git Flow / Feature Branching
 - main (production-ready)
 - develop (integration) – optional
 - feature/... untuk tiap fitur/bagian
 - 1.2. Naming: feature/<nama>, fix/<nama>, hotfix/<nama>
2. Issue Tracking
 - GitHub Issues / GitLab Issues
 - Buat issue per tugas: judul jelas, deskripsi, label (bug, enhancement, docs), assign ke anggota
 - Workflow: To Do → In Progress → Done (via Projects Board)
3. Pull Request (PR)
 - 3.1. Tujuan: code review, diskusi, tracing changes
 - 3.2. Gunakan PR template:
 - Deskripsi perubahan (What/Why)
 - Referensi Issue (#NN)
 - Checklist (tested, docs updated, coding standard)
4. Code Review
 - 4.1. Peer review: baca kode, jalankan, berikan komentar
 - 4.2. Fokus: correctness, style, security, performance, readability
 - 4.3. Checklist minimal:
 - Apakah fungsionalitas terpenuhi?
 - Apakah terdapat error handling & validasi?
 - Apakah naming & struktur folder konsisten?

- Apakah ada dokumentasi & test (jika perlu)?
5. Branch Protection & Merge Policy
 - Protect main agar hanya dapat di-merge via PR yang sudah di-approve minimal 1 reviewer
 - Wajib passing CI/tests & review sebelum merge

Praktikum

1. Pastikan Anda switch dari branch main ke branch main:

```
cd C:\laragon\www\inventory
git checkout main
git pull origin main
```

2. Siapkan PR & Issue Template
(Hanya perlu dilakukan sekali di repo)

- 2.1. Buat direktori .github:

```
mkdir -p .github
```

- 2.2. File .github/ISSUE_TEMPLATE.md:

```
## Judul
(singkat, jelas)

## Deskripsi
Jelaskan perubahan / bug / enhancement

## Langkah Reproduksi (jika bug)
1. ...
2. ...

## Harapan
(apa yang seharusnya terjadi)

## Label
- bug / enhancement / docs
```

- 2.3. File .github/PULL_REQUEST_TEMPLATE.md:

```
## What does this PR do?
- ...

## Related Issue
Closes #<issue-number>

## Changes
- ...

## Checklist
- [ ] I tested my code
- [ ] I updated documentation
- [ ] Code complies with coding standards
```

2.4. Commit & push template:

```
git add .
git commit -m "Add GitHub issue & PR templates"
git push -u origin main
```

3. Buat GitHub Issue

3.1. Buka halaman GitHub repo → Issues → New issue (pakai template)

Title: "Add filter items by category"

Label: enhancement

Assignee: anggota A

3.2. Klik Submit new issue, tercatat #NN.

4. Branching & Implementasi Fitur

4.1. Buat branch feature berdasarkan Issue:

```
git checkout -b feature/item-filter
```

4.2. Update ItemController@index untuk menerima query param category_id:

File: app/Http/Controllers/ItemController.php

```
<?php
namespace App\Http\Controllers;
use Illuminate\Http\Request;
use App\Services\ItemService;
use App\Http\Controllers\Api\BaseController;
class ItemController extends BaseController {
    // ... konstruktor ...
    public function index(Request $request){
        $items = $this->svc
            ->all()
            ->filter(fn($item) =>
                !$request->category_id
                || $item->category_id == $request->category_id
            );
        return $this->success($items);
    } // ... method lain ...
}
```

Atau, jika ingin di query level:

```
public function index(Request $request) {
    $query = Item::with('category');
    if ($request->filled('category_id')) {
        $query->where('category_id', $request->category_id);
    }
    return $this->success($query->get());
}
```

4.3. Update dokumentasi di [docs/api.md](#): tambahkan keterangan

GET /items?category_id={id}

Description: Filter items by category, optional.

5. Commit & Push Branch

```
git add .  
git commit -m "Feature: filter items by category (closes  
#NN)"  
git push -u origin feature/item-filter
```

6. Buka Pull Request

6.1. Di GitHub → Pull requests → New pull request → pilih branch
feature/item-filter

6.2. Title: "Feature: filter items by category"

6.3. Body: isi sesuai [.github/PULL_REQUEST_TEMPLATE.md](#)

- What does this PR do? "Menambahkan filter items by category pada endpoint GET /items"
- Related Issue: Closes #NN
- Changes: "Mengubah ItemController@index, memperbarui docs/api.md"
- Checklist: centang setelah testing manual

7. Code Review Peer

7.1. Reviewer A/B meninjau PR, komentar kalau ada yang perlu diperbaiki

7.2. Jika ada request changes, pembuat PR melakukan amend & push lagi ke branch yang sama

```
git add .  
git commit --amend # atau commit baru  
git push --force
```

8. Merge PR & Tutup Issue

- Setelah minimal 1 approval & semua checklist terpenuhi → klik "Merge pull request"
- Tambahkan commit message default ("Merge feature/item-filter...")
- Issue #NN akan otomatis ditutup karena ada "Closes #NN"